MUSAWENKOSI DUBE
Electrical Engineering Consultant
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SUMMARY

Software and Electronics Engineer with experience in electrical consulting, software development, mobile application development, web technologies, and database design. Skilled in developing cross-platform mobile applications using React Native and modern web applications using React.js, JavaScript, and TypeScript. Experienced in designing and deploying database-driven solutions using PostgreSQL, MongoDB, SQL, and SQLite. Strong programming background in Python, C#, Java, and C++, with hands-on experience using Git, Linux, REST APIs, and cloud-based technologies. Proven ability to combine engineering problem-solving, software development, and technical design to deliver practical and scalable solutions.

KEY COMPETENCIES

- Mobile Application Development
- Web Application Development
- Database Design & Management
- Software Architecture
- API Integration
- AI Chatbot Integration
- Full-Stack Development
- Problem Solving
- System Design
- Technical Documentation
- Engineering Design
- Project Coordination

EXPERIENCE

B2A Consulting engineering, Cape Town, South Africa

Electrical Engineering Consultant (06/2025 - Present)

- Involved in the design and coordination of electrical building services for commercial, residential, railway, and infrastructure projects.
- Performed electrical load calculations, cable sizing, voltage drop calculations, fault level assessments, and protection device selection for low-voltage electrical systems.
- Developed electrical designs for power reticulation, lighting, small power, distribution boards, substations, and building services infrastructure.
- Prepared electrical layouts, schematics, single-line diagrams, and detailed engineering drawings using AutoCAD.

- Assisted with lighting designs, DALI lighting control systems, emergency lighting systems, and energy-efficient electrical solutions.
- Designed and coordinated ICT, fibre-optic, access control, CCTV, and other extra-low-voltage systems.
- Prepared technical specifications, equipment schedules, bills of quantities, and engineering reports for tender and construction documentation.
- Conducted site inspections and construction monitoring to verify compliance with project specifications, standards, and quality requirements.
- Coordinated design activities with architects, structural engineers, mechanical engineers, contractors, and clients throughout project lifecycles.
- Assisted in contractor coordination, technical evaluations, project planning, and cost estimation activities.
- Investigated engineering issues, reviewed contractor submissions, and provided technical recommendations to support project delivery.
- Supported the implementation of electrical infrastructure upgrades, refurbishments, and new developments from concept design through construction.

EFSOL Russia, Saint Petersburg, Russia

Control Systems Engineer (07/2023 - 01/2025)

- Developed and implemented control-system solutions to address technical and operational requirements.
- Performed troubleshooting and maintenance activities to identify faults and improve system reliability.
- Analysed complex engineering problems and developed practical solutions while considering multiple technical constraints.
- Designed user interfaces for control systems to improve system operation and usability.

- Supported the ongoing improvement and reliable operation of electronic and control systems.

Svetlana X-Ray, Saint Petersburg, Russia

Electronics Engineer Intern (06/2021 - 09/2021)

- Assisted in the design of electronic components using KOMPAS-3D.
- Participated in the development and production of X-ray tubes and associated electronic systems.
- Supported engineering processes involving electronic equipment development, testing, and technical problem-solving.
- Documented engineering procedures and contributed to refining technical processes.
- Participated in technical training to strengthen engineering knowledge and understanding of industry practices.

Svetlana X-Ray, Saint Petersburg, Russia

Freelancing (06/2020 - Present)

- Developed full-stack mobile and web applications.
- Designed and managed cloud-connected database solutions.
- Implemented version control using Git and GitHub.
- Worked extensively within Linux-based development environments.
- Built API-driven applications integrating external services.
- Applied Agile development practices and iterative software delivery methods.

EDUCATION

Saint Petersburg Electrotechnical University, Russia

Master of Science (MSc) in Electronics and Nanotechnology Engineering (09/2022 - 09/2024)

- Specialization: Renewable Solar Energy
- **Dissertation:** *"Computer Simulation of Wireless Power Transmission Systems Using Lasers and Solar Cells"*

Bachelor of Science (BSc) in Electronics and Nanotechnology Engineering (09/2018 - 09/2022)

- Specialization: Electronic Devices and Applications
- **Dissertation:** *"Software Package for Modelling X-ray Image Parameters"*

Sizane High School, Bulawayo, Zimbabwe

- **A-Level:** 4 As
- **O-Level:** 8 As, 2 Bs, 1 Distinction

TECHNICAL SKILLS

Software Development: JavaScript , TypeScript, Python, C#, Java, C++, React.js, React Native, HTML, CSS, Kotlin and Swift.

Databases: PostgreSQL, MongoDB, SQL Server, SQLite, Database Design, Query Optimization.

Tools & Platforms: Git, GitHub, Linux, VS Code, REST APIs, JSON, Agile Development, Kubernetes and Docker.

Engineering Software: AutoCAD, Revit MEP, Dialux Evo, Relux, ETAP, Matlab, Simulink and Comsol Multiphysics, CODESYS, LabView, Gazebo simulator

Control Systems & Equipment Reliability

- Control systems design and troubleshooting
- Fault diagnosis and problem-solving
- Equipment reliability and maintenance support
- Embedded systems and programmable logic controllers

- Electronic equipment development and testing
- Arduino and STM32 programming

SOFTWARE DEVELOPMENT PROJECTS

Rent Management Mobile Application with AI Chatbot

- Designed and developed a mobile property rental management application using React Native.
- Integrated an AI chatbot to assist tenants and property managers with enquiries and support functions.
- Developed user authentication, property listing, messaging, and booking functionality.
- Implemented backend data storage and database integration.
- Optimized application performance across Android devices.

Global Airtime Sales Platform

- Designed and developed a mobile property rental management application using React Native.
- Integrated an AI chatbot to assist tenants and property managers with enquiries and support functions.
- Developed user authentication, property listing, messaging, and booking functionality.
- Implemented backend data storage and database integration.
- Optimized application performance across Android devices.

CERTIFICATIONS & TRAINING

- **Training Practice (4th Semester, BSc)** - Excellent
- **Field Experience (6th Semester, BSc)** - Excellent
- **Pre-Diploma Practical Training (8th Semester, BSc)** - Excellent
- **Academic Internship (1st Semester, MSc)** - Excellent
- **Internship (Research Project, MSc, 2nd-4th Semester)** – Excellent
- **Computer Vision (Academic programme)**: Hard and soft

LANGUAGES

English, **Ndebele**, **Shona** ,**Zulu**, **Russian**

PROFESSIONAL MEMBERSHIPS

Engineering Council of South Africa (ECSA) – Candidate Engineer

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